

Webscraping

[Numbeo](https://www.numbeo.com/cost-of-living/), is a website where users supply price information in different locations. The task is to get the cost of living in USD(\$) for all countries. Get the name of each product category, the price of the product in USD. Also get the cost of living relative to the United States. Save this together with the country names in a csv file

1. Use the request module to download and save as .htm <https://www.numbeo.com/cost-of-living/> .

```
In [15]: import requests
from bs4 import BeautifulSoup
import pandas as pd
import re
import time
import os

# Step 1: Download and save the main cost of living page
base_url = "https://www.numbeo.com/cost-of-living/"
response = requests.get(base_url)

# Save the main page as HTML
main_page_path = "numbeo_main_page.htm"
with open(main_page_path, "w", encoding="utf-8") as file:
    file.write(response.text)
```

2. Load the file into a BeautifulSoup object. Find all possible country input values. You can see all parameters in the URL, for example, https://www.numbeo.com/cost-of-living/country_result.jsp?country=Sweden&displayCurrency=USD. This means that you can just loop over country names in the url.

```
In [16]: # Step 2: Load the file into BeautifulSoup and find all country input values
with open(main_page_path, "r", encoding="utf-8") as file:
    soup = BeautifulSoup(file, "html.parser")

# Extract all countries from dropdown
countries = [option.get_text(strip=True) for option in soup.select("select[name='country']")]
print("Countries found:", countries)
# Extract all countries from the dropdown menu, filtering out placeholder options
countries = [option.get_text(strip=True) for option in soup.select("select[name='country']")]
print("Filtered Countries:", countries)

# Base URL for country-specific pages
country_base_url = "https://www.numbeo.com/cost-of-living/country_result.jsp?country={}&dis

# Directory to save country files
output_dir = "country_data"
os.makedirs(output_dir, exist_ok=True)

# Initialize list to store data for all countries
all_country_data = []
```

Countries found: ['---Select country---', 'Afghanistan', 'Aland Islands', 'Albania', 'Aldern ey', 'Algeria', 'American Samoa', 'Andorra', 'Angola', 'Anguilla', 'Antigua And Barbuda', 'A rgentina', 'Armenia', 'Aruba', 'Australia', 'Austria', 'Azerbaijan', 'Bahamas', 'Bahrain', 'Bangladesh', 'Barbados', 'Belarus', 'Belgium', 'Belize', 'Benin', 'Bermuda', 'Bhutan', 'Bol ivia', 'Bonaire', 'Bosnia And Herzegovina', 'Botswana', 'Brazil', 'British Virgin Islands', 'Brunei', 'Bulgaria', 'Burkina Faso', 'Burundi', 'Cambodia', 'Cameroon', 'Canada', 'Cape Ver de', 'Cayman Islands', 'Central African Republic', 'Chad', 'Chile', 'China', 'Colombia', 'Co moros', 'Cook Islands', 'Costa Rica', 'Croatia', 'Cuba', 'Curacao', 'Cyprus', 'Czech Republi c', 'Democratic Republic of the Congo', 'Denmark', 'Djibouti', 'Dominica', 'Dominican Republ ic', 'Ecuador', 'Egypt', 'El Salvador', 'Equatorial Guinea', 'Eritrea', 'Estonia', 'Eswatin i', 'Ethiopia', 'Falkland Islands', 'Faroe Islands', 'Fiji', 'Finland', 'France', 'French Gu iana', 'French Polynesia', 'French Southern Territories', 'Gabon', 'Gambia', 'Georgia', 'Ger many', 'Ghana', 'Gibraltar', 'Greece', 'Greenland', 'Grenada', 'Guadeloupe', 'Guam', 'Guatem ala', 'Guernsey', 'Guinea', 'Guinea-Bissau', 'Guyana', 'Haiti', 'Honduras', 'Hong Kong (Chin a)', 'Hungary', 'Iceland', 'India', 'Indonesia', 'Iran', 'Iraq', 'Ireland', 'Isle Of Man', 'Israel', 'Italy', 'Ivory Coast', 'Jamaica', 'Japan', 'Jersey', 'Jordan', 'Kazakhstan', 'Ken ya', 'Kiribati', 'Kosovo (Disputed Territory)', 'Kuwait', 'Kyrgyzstan', 'Laos', 'Latvia', 'L ebanon', 'Lesotho', 'Liberia', 'Libya', 'Liechtenstein', 'Lithuania', 'Luxembourg', 'Macao (China)', 'Madagascar', 'Malawi', 'Malaysia', 'Maldives', 'Mali', 'Malta', 'Marshall Island s', 'Martinique', 'Mauritania', 'Mauritius', 'Mexico', 'Micronesia', 'Moldova', 'Monaco', 'M ongolia', 'Montenegro', 'Montserrat', 'Morocco', 'Mozambique', 'Myanmar', 'Namibia', 'Naur u', 'Nepal', 'Netherlands', 'New Caledonia', 'New Zealand', 'Nicaragua', 'Niger', 'Nigeria', 'Norfolk Island', 'North Korea', 'North Macedonia', 'Northern Mariana Islands', 'Norway', 'O man', 'Pakistan', 'Palau', 'Palestine', 'Panama', 'Papua New Guinea', 'Paraguay', 'Peru', 'P hilippines', 'Poland', 'Portugal', 'Puerto Rico', 'Qatar', 'Republic of the Congo', 'Reunio n', 'Romania', 'Russia', 'Rwanda', 'Saint Helena', 'Saint Kitts And Nevis', 'Saint Lucia', 'Saint Vincent And The Grenadines', 'Saint-Pierre And Miquelon', 'Samoa', 'San Marino', 'Sao Tome And Principe', 'Saudi Arabia', 'Senegal', 'Serbia', 'Seychelles', 'Sierra Leone', 'Sing apore', 'Sint Maarten', 'Slovakia', 'Slovenia', 'Solomon Islands', 'Somalia', 'South Afric a', 'South Korea', 'South Sudan', 'Spain', 'Sri Lanka', 'Sudan', 'Suriname', 'Sweden', 'Swit zerland', 'Syria', 'Taiwan', 'Tajikistan', 'Tanzania', 'Thailand', 'Timor-Leste', 'Togo', 'T onga', 'Trinidad And Tobago', 'Tunisia', 'Turkey', 'Turkmenistan', 'Turks And Caicos Island s', 'Tuvalu', 'Uganda', 'Ukraine', 'United Arab Emirates', 'United Kingdom', 'United State s', 'Uruguay', 'Us Virgin Islands', 'Uzbekistan', 'Vanuatu', 'Venezuela', 'Vietnam', 'Wallis And Futuna', 'Western Sahara', 'Yemen', 'Zambia', 'Zimbabwe']

Filtered Countries: ['Afghanistan', 'Aland Islands', 'Albania', 'Alderney', 'Algeria', 'Amer ican Samoa', 'Andorra', 'Angola', 'Anguilla', 'Antigua And Barbuda', 'Argentina', 'Armenia', 'Aruba', 'Australia', 'Austria', 'Azerbaijan', 'Bahamas', 'Bahrain', 'Bangladesh', 'Barbado s', 'Belarus', 'Belgium', 'Belize', 'Benin', 'Bermuda', 'Bhutan', 'Bolivia', 'Bonaire', 'Bos nia And Herzegovina', 'Botswana', 'Brazil', 'British Virgin Islands', 'Brunei', 'Bulgaria', 'Burkina Faso', 'Burundi', 'Cambodia', 'Cameroon', 'Canada', 'Cape Verde', 'Cayman Islands', 'Central African Republic', 'Chad', 'Chile', 'China', 'Colombia', 'Comoros', 'Cook Islands', 'Costa Rica', 'Croatia', 'Cuba', 'Curacao', 'Cyprus', 'Czech Republic', 'Democratic Republic of the Congo', 'Denmark', 'Djibouti', 'Dominica', 'Dominican Republic', 'Ecuador', 'Egypt', 'El Salvador', 'Equatorial Guinea', 'Eritrea', 'Estonia', 'Eswatini', 'Ethiopia', 'Falkland Islands', 'Faroe Islands', 'Fiji', 'Finland', 'France', 'French Guiana', 'French Polynesia', 'French Southern Territories', 'Gabon', 'Gambia', 'Georgia', 'Germany', 'Ghana', 'Gibralta r', 'Greece', 'Greenland', 'Grenada', 'Guadeloupe', 'Guam', 'Guatemala', 'Guernsey', 'Guine a', 'Guinea-Bissau', 'Guyana', 'Haiti', 'Honduras', 'Hong Kong (China)', 'Hungary', 'Icelan d', 'India', 'Indonesia', 'Iran', 'Iraq', 'Ireland', 'Isle Of Man', 'Israel', 'Italy', 'Ivor y Coast', 'Jamaica', 'Japan', 'Jersey', 'Jordan', 'Kazakhstan', 'Kenya', 'Kiribati', 'Kosovo (Disputed Territory)', 'Kuwait', 'Kyrgyzstan', 'Laos', 'Latvia', 'Lebanon', 'Lesotho', 'Libe ria', 'Libya', 'Liechtenstein', 'Lithuania', 'Luxembourg', 'Macao (China)', 'Madagascar', 'M alawi', 'Malaysia', 'Maldives', 'Mali', 'Malta', 'Marshall Islands', 'Martinique', 'Mauritan ia', 'Mauritius', 'Mexico', 'Micronesia', 'Moldova', 'Monaco', 'Mongolia', 'Montenegro', 'Mo ntserrat', 'Morocco', 'Mozambique', 'Myanmar', 'Namibia', 'Nauru', 'Nepal', 'Netherlands', 'New Caledonia', 'New Zealand', 'Nicaragua', 'Niger', 'Nigeria', 'Norfolk Island', 'North Ko rea', 'North Macedonia', 'Northern Mariana Islands', 'Norway', 'Oman', 'Pakistan', 'Palau', 'Palestine', 'Panama', 'Papua New Guinea', 'Paraguay', 'Peru', 'Philippines', 'Poland', 'Por tugal', 'Puerto Rico', 'Qatar', 'Republic of the Congo', 'Reunion', 'Romania', 'Russia', 'Rw anda', 'Saint Helena', 'Saint Kitts And Nevis', 'Saint Lucia', 'Saint Vincent And The Grenad ines', 'Saint-Pierre And Miquelon', 'Samoa', 'San Marino', 'Sao Tome And Principe', 'Saudi A rabia', 'Senegal', 'Serbia', 'Seychelles', 'Sierra Leone', 'Singapore', 'Sint Maarten', 'Slo vakia', 'Slovenia', 'Solomon Islands', 'Somalia', 'South Africa', 'South Korea', 'South Suda n', 'Spain', 'Sri Lanka', 'Sudan', 'Suriname', 'Sweden', 'Switzerland', 'Syria', 'Taiwan', 'Tajikistan', 'Tanzania', 'Thailand', 'Timor-Leste', 'Togo', 'Tonga', 'Trinidad And Tobago', 'Tunisia', 'Turkey', 'Turkmenistan', 'Turks And Caicos Islands', 'Tuvalu', 'Uganda', 'Ukrain

e', 'United Arab Emirates', 'United Kingdom', 'United States', 'Uruguay', 'Us Virgin Islands', 'Uzbekistan', 'Vanuatu', 'Venezuela', 'Vietnam', 'Wallis And Futuna', 'Western Sahara', 'Yemen', 'Zambia', 'Zimbabwe']

3. Download and save one country file. Load the file into a BeautifulSoup object. Find the relevant table. Print the table headings and values, together with the country name to a file. Then use regular expressions to extract the cost of living relative to the Sweden. Add this to the printed table.

```
In [17]: #Step 3: Test with Czech Republic
for country in countries:
    if country != "Czech Republic":
        continue # Skip all countries except Czech Republic

    # Format the URL
    country_url = country_base_url.format(country.replace(" ", "+"))
    country_response = requests.get(country_url)

    # Save the Czech Republic page as an HTML file
    country_file_path = os.path.join(output_dir, f"{country}_cost_of_living.htm")
    with open(country_file_path, "w", encoding="utf-8") as file:
        file.write(country_response.text)

    # Parse the country file
    country_soup = BeautifulSoup(country_response.content, "html.parser")
    table = country_soup.find("table", {"class": "data_wide_table"})

    # Extract table data for Czech Republic
    country_data = []
    if table:
        headers = [th.get_text(strip=True) for th in table.find_all("th") if th.get_text(strip=True)]
        for row in table.find_all("tr"):
            cells = row.find_all("td")
            if len(cells) >= 2:
                category = cells[0].get_text(strip=True)
                price = cells[1].get_text(strip=True)
                country_data.append({"Country": country, "Category": category, "Price (USD)": price})

    # Extract cost of living index relative to Sweden
    summary_section = country_soup.find("div", class_="seeding-call table_color summary_line")
    relative_cost_text = "N/A"
    if summary_section:
        relative_cost_match = re.search(r"Cost of living in.*is, on average, (\d+\.\d+)%\s*", summary_section.get_text())
        if relative_cost_match:
            relative_cost_text = f"{relative_cost_match.group(1)} {relative_cost_match.group(2)}%"

    # Add relative cost info to each entry
    for entry in country_data:
        entry["Relative Cost to Sweden"] = relative_cost_text

print(country_data)
```

[{'Country': 'Czech Republic', 'Category': 'Meal, Inexpensive Restaurant', 'Price (USD)': '8.53\xa0\$', 'Relative Cost to Sweden': '7.2% lower'}, {'Country': 'Czech Republic', 'Category': 'Meal for 2 People, Mid-range Restaurant, Three-course', 'Price (USD)': '42.64\xa0\$', 'Relative Cost to Sweden': '7.2% lower'}, {'Country': 'Czech Republic', 'Category': 'McMeal at McDonalds (or Equivalent Combo Meal)', 'Price (USD)': '7.89\xa0\$', 'Relative Cost to Sweden': '7.2% lower'}, {'Country': 'Czech Republic', 'Category': 'Domestic Beer (0.5 liter draught)', 'Price (USD)': '2.13\xa0\$', 'Relative Cost to Sweden': '7.2% lower'}, {'Country': 'Czech Republic', 'Category': 'Imported Beer (0.33 liter bottle)', 'Price (USD)': '2.56\xa0\$', 'Relative Cost to Sweden': '7.2% lower'}, {'Country': 'Czech Republic', 'Category': 'Cappuccino (regular)', 'Price (USD)': '2.78\xa0\$', 'Relative Cost to Sweden': '7.2% lower'}, {'Country': 'Czech Republic', 'Category': 'Coke/Pepsi (0.33 liter bottle)', 'Price (USD)': '1.6\xa0\$', 'Relative Cost to Sweden': '7.2% lower'}, {'Country': 'Czech Republic', 'Category': 'Water (0.33 liter bottle)', 'Price (USD)': '1.32\xa0\$', 'Relative Cost to Sweden': '7.2% lower'}, {'Country': 'Czech Republic', 'Category': 'Milk (regular), (1 liter)', 'Price (USD)': '1.03\xa0\$', 'Relative Cost to Sweden': '7.2% lower'}, {'Country': 'Czech Republic', 'Category': 'Loaf of Fresh White Bread (500g)', 'Price (USD)': '1.41\xa0\$', 'Relative Cost to Sweden': '7.2% lower'}, {'Country': 'Czech Republic', 'Category': 'Rice (white), (1kg)', 'Price (USD)': '2.22\xa0\$', 'Relative Cost to Sweden': '7.2% lower'}, {'Country': 'Czech Republic', 'Category': 'Eggs (regular) (12)', 'Price (USD)': '2.68\xa0\$', 'Relative Cost to Sweden': '7.2% lower'}, {'Country': 'Czech Republic', 'Category': 'Local Cheese (1kg)', 'Price (USD)': '11.26\xa0\$', 'Relative Cost to Sweden': '7.2% lower'}, {'Country': 'Czech Republic', 'Category': 'Chicken Fillets (1kg)', 'Price (USD)': '8.51\xa0\$', 'Relative Cost to Sweden': '7.2% lower'}, {'Country': 'Czech Republic', 'Category': 'Beef Round (1kg) (or Equivalent Back Leg Red Meat)', 'Price (USD)': '12.03\xa0\$', 'Relative Cost to Sweden': '7.2% lower'}, {'Country': 'Czech Republic', 'Category': 'Apples (1kg)', 'Price (USD)': '1.51\xa0\$', 'Relative Cost to Sweden': '7.2% lower'}, {'Country': 'Czech Republic', 'Category': 'Banana (1kg)', 'Price (USD)': '1.62\xa0\$', 'Relative Cost to Sweden': '7.2% lower'}, {'Country': 'Czech Republic', 'Category': 'Oranges (1kg)', 'Price (USD)': '2.04\xa0\$', 'Relative Cost to Sweden': '7.2% lower'}, {'Country': 'Czech Republic', 'Category': 'Tomato (1kg)', 'Price (USD)': '3.0\xa0\$', 'Relative Cost to Sweden': '7.2% lower'}, {'Country': 'Czech Republic', 'Category': 'Potato (1kg)', 'Price (USD)': '1.16\xa0\$', 'Relative Cost to Sweden': '7.2% lower'}, {'Country': 'Czech Republic', 'Category': 'Onion (1kg)', 'Price (USD)': '1.13\xa0\$', 'Relative Cost to Sweden': '7.2% lower'}, {'Country': 'Czech Republic', 'Category': 'Lettuce (1 head)', 'Price (USD)': '1.35\xa0\$', 'Relative Cost to Sweden': '7.2% lower'}, {'Country': 'Czech Republic', 'Category': 'Water (1.5 liter bottle)', 'Price (USD)': '0.69\xa0\$', 'Relative Cost to Sweden': '7.2% lower'}, {'Country': 'Czech Republic', 'Category': 'Bottle of Wine (Mid-Range)', 'Price (USD)': '6.18\xa0\$', 'Relative Cost to Sweden': '7.2% lower'}, {'Country': 'Czech Republic', 'Category': 'Domestic Beer (0.5 liter bottle)', 'Price (USD)': '0.86\xa0\$', 'Relative Cost to Sweden': '7.2% lower'}, {'Country': 'Czech Republic', 'Category': 'Imported Beer (0.33 liter bottle)', 'Price (USD)': '1.82\xa0\$', 'Relative Cost to Sweden': '7.2% lower'}, {'Country': 'Czech Republic', 'Category': 'Cigarettes 20 Pack (Marlboro)', 'Price (USD)': '6.4\xa0\$', 'Relative Cost to Sweden': '7.2% lower'}, {'Country': 'Czech Republic', 'Category': 'One-way Ticket (Local Transport)', 'Price (USD)': '1.28\xa0\$', 'Relative Cost to Sweden': '7.2% lower'}, {'Country': 'Czech Republic', 'Category': 'Monthly Pass (Regular Price)', 'Price (USD)': '23.45\xa0\$', 'Relative Cost to Sweden': '7.2% lower'}, {'Country': 'Czech Republic', 'Category': 'Taxi Start (Normal Tariff)', 'Price (USD)': '2.13\xa0\$', 'Relative Cost to Sweden': '7.2% lower'}, {'Country': 'Czech Republic', 'Category': 'Taxi 1km (Normal Tariff)', 'Price (USD)': '1.28\xa0\$', 'Relative Cost to Sweden': '7.2% lower'}, {'Country': 'Czech Republic', 'Category': 'Taxi 1hour Waiting (Normal Tariff)', 'Price (USD)': '15.35\xa0\$', 'Relative Cost to Sweden': '7.2% lower'}, {'Country': 'Czech Republic', 'Category': 'Gasoline (1 liter)', 'Price (USD)': '1.6\xa0\$', 'Relative Cost to Sweden': '7.2% lower'}, {'Country': 'Czech Republic', 'Category': 'Volkswagen Golf 1.4 90 KW Trendline (Or Equivalent New Car)', 'Price (USD)': '25,585.7\xa0\$', 'Relative Cost to Sweden': '7.2% lower'}, {'Country': 'Czech Republic', 'Category': 'Toyota Corolla Sedan 1.6l 97kW Comfort (Or Equivalent New Car)', 'Price (USD)': '27,185.89\xa0\$', 'Relative Cost to Sweden': '7.2% lower'}, {'Country': 'Czech Republic', 'Category': 'Basic (Electricity, Heating, Cooling, Water, Garbage) for 85m2 Apartment', 'Price (USD)': '283.6\xa0\$', 'Relative Cost to Sweden': '7.2% lower'}, {'Country': 'Czech Republic', 'Category': 'Mobile Phone Monthly Plan with Calls and 10GB+ Data', 'Price (USD)': '27.13\xa0\$', 'Relative Cost to Sweden': '7.2% lower'}, {'Country': 'Czech Republic', 'Category': 'Internet (60 Mbps or More, Unlimited Data, Cable/ADSL)', 'Price (USD)': '19.76\xa0\$', 'Relative Cost to Sweden': '7.2% lower'}, {'Country': 'Czech Republic', 'Category': 'Fitness Club, Monthly Fee for 1 Adult', 'Price (USD)': '54.53\xa0\$', 'Relative Cost to Sweden': '7.2% lower'}, {'Country': 'Czech Republic', 'Category': 'Tennis Court Rent (1 Hour on Weekend)', 'Price (USD)': '15.54\xa0\$', 'Relative Cost to Sweden': '7.2% lower'}, {'Country': 'Czech Republic', 'Category': 'Cinema, International Release, 1 Seat', 'Price (USD)': '9.81\xa0\$', 'Relative Cost to Sweden': '7.2% lower'}, {'Country': 'Czech Republic', 'Category': 'Preschool (or Kindergarten), Full Day, Private, Monthly for 1 Chi

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ld', 'Price (USD)': '463.93\xa0$', 'Relative Cost to Sweden': '7.2% lower'}, {'Country': 'Czech Republic', 'Category': 'International Primary School, Yearly for 1 Child', 'Price (USD)': '10,070.46\xa0$', 'Relative Cost to Sweden': '7.2% lower'}, {'Country': 'Czech Republic', 'Category': '1 Pair of Jeans (Levis 501 Or Similar)', 'Price (USD)': '78.45\xa0$', 'Relative Cost to Sweden': '7.2% lower'}, {'Country': 'Czech Republic', 'Category': '1 Summer Dress in a Chain Store (Zara, H&M, ...)', 'Price (USD)': '37.95\xa0$', 'Relative Cost to Sweden': '7.2% lower'}, {'Country': 'Czech Republic', 'Category': '1 Pair of Nike Running Shoes (Mid-Range)', 'Price (USD)': '89.81\xa0$', 'Relative Cost to Sweden': '7.2% lower'}, {'Country': 'Czech Republic', 'Category': '1 Pair of Men Leather Business Shoes', 'Price (USD)': '107.22\xa0$', 'Relative Cost to Sweden': '7.2% lower'}, {'Country': 'Czech Republic', 'Category': 'Apartment (1 bedroom) in City Centre', 'Price (USD)': '814.77\xa0$', 'Relative Cost to Sweden': '7.2% lower'}, {'Country': 'Czech Republic', 'Category': 'Apartment (1 bedroom) Outside of Centre', 'Price (USD)': '650.77\xa0$', 'Relative Cost to Sweden': '7.2% lower'}, {'Country': 'Czech Republic', 'Category': 'Apartment (3 bedrooms) in City Centre', 'Price (USD)': '1,405.55\xa0$', 'Relative Cost to Sweden': '7.2% lower'}, {'Country': 'Czech Republic', 'Category': 'Apartment (3 bedrooms) Outside of Centre', 'Price (USD)': '1,058.14\xa0$', 'Relative Cost to Sweden': '7.2% lower'}, {'Country': 'Czech Republic', 'Category': 'Price per Square Meter to Buy Apartment in City Centre', 'Price (USD)': '4,730.09\xa0$', 'Relative Cost to Sweden': '7.2% lower'}, {'Country': 'Czech Republic', 'Category': 'Price per Square Meter to Buy Apartment Outside of Centre', 'Price (USD)': '3,545.67\xa0$', 'Relative Cost to Sweden': '7.2% lower'}, {'Country': 'Czech Republic', 'Category': 'Average Monthly Net Salary (After Tax)', 'Price (USD)': '1,559.52\xa0$', 'Relative Cost to Sweden': '7.2% lower'}, {'Country': 'Czech Republic', 'Category': 'Mortgage Interest Rate in Percentages (%), Yearly, for 20 Years Fixed-Rate', 'Price (USD)': '5.68', 'Relative Cost to Sweden': '7.2% lower'}]]
```

4. Loop over all countries and do step 3. Note that URLs never include whitespaces; therefore, check what happens when a country name consists of multiple words.

```
In [18]: # Step 4: Loop over each country, download page, and parse cost of living data
for country in countries:
    country_url = country_base_url.format(country.replace(" ", "+"))
    country_response = requests.get(country_url)

    # Save country page as HTML file
    country_file_path = os.path.join(output_dir, f"{country}_cost_of_living.htm")
    with open(country_file_path, "w", encoding="utf-8") as file:
        file.write(country_response.text)

    # Parse the country file
    country_soup = BeautifulSoup(country_response.content, "html.parser")
    table = country_soup.find("table", {"class": "data_wide_table"})

    # Extract table data
    country_data = []
    if table:
        headers = [th.get_text(strip=True) for th in table.find_all("th") if th.get_text(strip=True)]
        for row in table.find_all("tr"):
            cells = row.find_all("td")
            if len(cells) >= 2:
                category = cells[0].get_text(strip=True)
                price = cells[1].get_text(strip=True)
                country_data.append({"Country": country, "Category": category, "Price (USD)": price})

    # Extract cost of living index relative to Sweden
    summary_section = country_soup.find("div", class_="seeding-call table_color summary limit")
    relative_cost_text = "N/A"
    if summary_section:
        relative_cost_match = re.search(r"Cost of living in.*is, on average, (\d+\.\d+)%\s*", summary_section.get_text())
        if relative_cost_match:
            relative_cost_text = f"{relative_cost_match.group(1)} {relative_cost_match.group(2)}"

    # Add relative cost info to each entry for this country
    for entry in country_data:
```

```

entry["Relative Cost to Sweden"] = relative_cost_text

# Add this country's data to the all_country_data list
all_country_data.extend(country_data)

# Delay to avoid hitting the server too frequently
time.sleep(1)

```

5. Read all files into Stata

```

In [26]: # Step 5: Save all data to a CSV file
output_csv = "numbeo_cost_of_living_all_countries.csv"
df = pd.DataFrame(all_country_data)
df.head()

```

Out[26]:

	Country	Category	Price (USD)	Relative Cost to Sweden
0	Afghanistan	Meal, Inexpensive Restaurant	2.25 \$	87.5% lower
1	Afghanistan	Meal for 2 People, Mid-range Restaurant, Three...	9.5 \$	87.5% lower
2	Afghanistan	McMeal at McDonalds (or Equivalent Combo Meal)	3.75 \$	87.5% lower
3	Afghanistan	Domestic Non-Alcoholic Beer (0.5 liter draught)	3.0 \$	87.5% lower
4	Afghanistan	Imported Non-Alcoholic Beer (0.33 liter bottle)	7.5 \$	87.5% lower

```

In [27]: # Replace non-numeric values '?' with NaN, and remove commas and currency symbols for conversion
df['Price (USD)'] = df['Price (USD)'].replace('[\$,]', '', regex=True).replace('?', None).astype(float)
# Cleaning and reformatting Relative Cost to Sweden column
relative_cost = df['Relative Cost to Sweden'].str.extract(r'(\d+\.\d*)').astype(float)
sign = df['Relative Cost to Sweden'].str.contains("lower").map({True: -1, False: 1})
df['Relative Cost to Sweden'] = relative_cost[0] * sign

# Renaming the last column for clarity
df.rename(columns={'Relative Cost to Sweden': 'Relative Cost (%) to Sweden'}, inplace=True)

#save
df.to_csv(output_csv, index=False)

#showing the cleaned data
df.head()

```

Out[27]:

	Country	Category	Price (USD)	Relative Cost (%) to Sweden
0	Afghanistan	Meal, Inexpensive Restaurant	2.25	-87.5
1	Afghanistan	Meal for 2 People, Mid-range Restaurant, Three...	9.50	-87.5
2	Afghanistan	McMeal at McDonalds (or Equivalent Combo Meal)	3.75	-87.5
3	Afghanistan	Domestic Non-Alcoholic Beer (0.5 liter draught)	3.00	-87.5
4	Afghanistan	Imported Non-Alcoholic Beer (0.33 liter bottle)	7.50	-87.5